

ALEXANDER BAY, South Africa

WMO: 684030

Lat: 28.570S

Lon: 16.529E

Elev: 92

StdP: 14.65

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	42.9	44.9	23.2	17.6	71.5	28.1	22.1	69.8	28.3	60.6	25.6	61.2	3.4	150	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.6	89.6	62.9	84.4	61.4	79.8	61.1	69.0	78.4	67.5	75.2	66.4	73.5	11.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.1	96.4	71.8	64.7	92.0	70.5	63.8	88.8	69.7	33.2	78.5	32.0	75.2	31.1	73.6	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
31.7	27.5	23.6	DB	40.0	98.7	1.5	4.1	38.9	101.7	38.0	104.1	37.1	106.4	36.0	109.3	
			WB	37.5	71.4	1.3	2.2	36.5	73.0	35.8	74.2	35.0	75.5	34.0	77.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	63.0	67.8	67.8	66.2	64.6	61.7	59.4	58.9	57.9	59.4	61.6	64.1	66.8
	DBStd	5.74	2.92	3.11	4.11	6.11	5.45	5.68	6.49	4.71	4.50	3.57	3.02	2.78
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0
	HDD65	1258	6	6	28	74	136	191	219	235	182	117	50	14
	CDD50	4741	552	499	502	439	362	283	278	244	282	359	422	519
	CDD65	525	93	85	65	63	33	25	31	14	15	12	21	68
	CDH74	2764	169	173	307	556	321	237	293	171	220	129	107	81
	CDH80	1012	27	33	122	263	134	71	86	66	103	52	44	11
Wind	WSAvg	9.0	11.0	10.1	9.2	7.3	6.6	7.0	7.0	7.7	9.2	10.5	11.3	11.7
Precipitation	PrecAvg	1.7	0.0	0.1	0.1	0.2	0.2	0.3	0.2	0.2	0.2	0.2	0.0	0.1
	PrecMax	3.6	0.4	0.5	0.6	0.7	1.5	1.9	0.8	1.2	1.0	0.7	0.5	0.6
	PrecMin	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.8	0.1	0.1	0.1	0.2	0.3	0.4	0.2	0.3	0.2	0.2	0.1	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.3	84.7	91.8	96.6	91.1	87.2	87.3	88.2	91.4	88.6	86.0	80.0
		MCWB	65.2	69.0	66.6	66.7	61.8	60.1	58.9	58.2	64.2	62.2	61.2	64.1
	2%	DB	77.3	77.8	81.5	87.6	83.2	80.9	82.0	78.5	79.2	77.2	75.4	75.7
		MCWB	66.3	66.5	65.8	64.9	59.5	57.7	57.5	55.1	59.1	60.8	61.2	64.7
	5%	DB	75.2	75.5	75.5	79.0	75.1	74.6	76.2	70.0	70.9	71.1	71.9	73.8
		MCWB	65.5	65.8	65.0	62.7	57.7	56.1	55.9	53.4	56.8	59.8	61.5	64.0
	10%	DB	73.4	73.6	72.6	73.1	69.9	68.8	70.0	65.2	66.3	68.2	70.0	72.3
		MCWB	64.4	64.9	63.7	61.9	58.6	55.2	54.1	54.8	56.6	59.1	60.9	63.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	70.6	70.5	70.0	65.1	64.0	61.9	60.1	65.6	66.8	65.4	68.4
		MCDB	76.6	80.1	85.1	88.1	76.1	78.9	79.4	74.4	86.5	83.7	73.7	73.8
	2%	WB	68.2	68.2	67.8	67.2	62.5	60.4	59.4	58.2	60.1	62.8	63.7	66.5
		MCDB	74.2	74.9	76.8	80.6	72.2	71.2	73.7	68.5	72.6	71.1	70.5	72.4
	5%	WB	66.9	67.0	66.0	64.8	61.1	58.8	57.7	57.0	58.6	61.1	62.8	65.4
		MCDB	72.7	73.3	73.3	74.6	69.5	68.4	69.4	64.9	67.3	68.1	69.3	71.3
	10%	WB	65.6	65.9	64.6	62.9	59.9	57.4	56.3	56.0	57.5	60.0	61.9	64.5
		MCDB	71.3	71.9	71.1	71.2	67.4	65.6	65.7	62.9	64.6	66.4	68.2	70.4
Mean Daily Temperature Range	5% DB	MDBR	12.6	12.8	14.4	17.1	17.5	19.0	19.3	17.5	16.6	14.8	13.8	12.5
		MCDBR	14.3	14.8	20.4	29.3	29.0	29.2	29.6	30.7	28.9	22.1	17.3	14.0
		MCWBR	6.0	6.6	8.7	11.0	12.0	12.0	12.3	12.9	11.7	9.7	7.6	6.0
	5% WB	MCDBR	12.6	12.9	17.0	22.0	19.1	20.8	22.4	18.9	19.2	16.4	14.4	12.0
		MCWBR	6.1	6.1	7.7	9.5	9.6	11.1	11.5	9.9	9.3	8.1	7.2	5.7
Clear-Sky Solar Irradiance	taub		0.360	0.370	0.355	0.328	0.308	0.298	0.308	0.328	0.354	0.361	0.342	0.346
	taud		2.464	2.461	2.487	2.549	2.569	2.556	2.518	2.449	2.358	2.368	2.470	2.475
	Ebn at Noon		311	303	297	291	282	278	279	285	292	302	315	316
	Edh at Noon		38	37	34	29	26	24	26	31	38	40	37	38
All-Sky Solar Radiation	RadAvg		2568	2347	2026	1596	1249	1102	1181	1433	1835	2239	2569	2700
	RadStd		108	87	62	61	43	36	48	58	68	56	75	96

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page